

10-606 Recitation

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Warmup

Solve the following matrix equation for X : $(A^\top XB)^{-1} + C = D$

PSD Matrices and Inverses

1. Show that if A is invertible, then its eigenvalues are all nonzero. Conversely, if an eigenvalue is zero, why can't A^{-1} exist?
2. If A is invertible, verify that

$$(A + uv^\top)^{-1} = A^{-1} - \frac{A^{-1}u v^\top A^{-1}}{1 + v^\top A^{-1}u},$$

as long as the denominator is nonzero.

3. A *norm* is a function ρ that satisfies
 - (i) $\rho(x) \geq 0$ for all x
 - (ii) $\rho(x) = 0$ if and only if $x = 0$
 - (iii) $\rho(cx) = |c|\rho(x)$ for all x and scalars c
 - (iv) $\rho(x + y) \leq \rho(x) + \rho(y)$ for all x, y .

If A is a positive-definite matrix A , show that $\rho(x) = \|x\|_A$ where $\|x\|_A = \sqrt{x^\top A x}$ defines a norm. Recall that A is positive-definite if $x^\top A x > 0$ for all nonzero x . Hint: You may use the Cauchy-Schwarz inequality $x^\top A y \leq \|x\|_A \|y\|_A$ without proof.

SVD and Rank

1. What's the singular value decomposition of a matrix A ?
2. Show that if $A = uv^\top$ with $u \in \mathbb{R}^m$, $v \in \mathbb{R}^n$ and $u, v \neq 0$, then $\text{rank}(A) = 1$.
3. What is the relation between singular values of A and eigenvalues of $A^\top A$?
4. Compute the nonzero singular value of $A = uv^\top$.
5. If $A = U\Sigma V^\top$, what is the SVD of $A^\top$?